Home Reading 02 — Laplace Transform

Before Class

Preview Lecture 2

Read Lecture 2: Laplace Transform, and Lecture 02A: Partial Fractions (required self-study for Tutorial 2).

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Section 2-3, pp. 14–17 (Laplace transform and basic pairs).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.
- Reference only: Section 2-4, pp. 29–33, for inverse transforms; use Lecture 02A for the required partial-fraction method.

This Week Introduces

Laplace transforms, initial conditions, inverse transforms and checks using the initial-value and final-value theorems.

Tutorial Preparation

Read Tutorial 2 before class. Be ready to:

- Q1: Transform basic inputs and include nonzero initial conditions.
- Q2: Compare RC charging responses from two initial voltages.
- Q3: Distinguish repeated poles from complex poles.
- Q4: Decide whether the final-value theorem can be used.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- What does a Laplace transform represent, and why does it help solve a differential equation?
- Where does a nonzero initial condition appear after transforming an ODE?

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